

RESEARCH PAPER

An Additional L451G452N453 in the RpoB Protein Suppressed the Synthetic Lethality in *Escherichia coli* at 37 Degrees Caused by Depletion of DnaK/J and Trigger Factor

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ABSTRACT

Escherichia coli depletion of chaperone trigger factor and DnaK/J were not viable at 37°C, but viable below 30°C. Among the engineered *E. coli* depleted of trigger factor and DnaK/J, one strain Z625, exhibited survival at 37°C, while another strain Z629 only survived below 30°C. Comparative analysis of fatty acid profiles of Z625 and Z629 revealed absence of numerous saturated fatty acids in Z625 as compared to the wild-type *E. coli* BW25113. In addition, increased unsaturated fatty acids were present in Z625, whereas the fatty acids profile of Z629 closely resembled that of BW25113. Whole genome sequencing revealed a 9-bp insertion in *rpoB* of Z625. Combined structural analysis of simulated RpoB protein bearing the amino acid sequence L451G452N453 insertion and susceptibility analysis to rifampicin suggested that the insertion did not disturb the individual RpoB structure as beta subunit of RNA polymerase. Comparative transcriptomic analysis of Z625 and Z629 suggested that this insertion impacted transcription of the overall RNA polymerase in Z625, leading to potential repression of some genes whose overexpression was toxic to *E. coli*. Additionally, Z625 exhibited distinctive metabolic adaptations, likely contributing to its survival at 37°C. In summary, our study elucidated one LGN insertion in *rpoB* that impacts transcriptional regulation in *E. coli*, thereby explaining the survival of *E. coli* depletion of trigger factor and DnaK/J at 37°C, and these findings suggested that some simple mutations in critical genes like *rpoB* might play an important role in driving adaptive evolution.

Abbreviations: CDS, coding sequences; *chl*, chloramphenicol resistance gene; DEG, differential expressed gene; FA, fatty acid; FAME, fatty acid methyl ester; GO, gene ontology; I-TASSER, Iterative Threading Assembly Refinement; *kan*, kanamycin resistance gene; KEGG, Kyoto Encyclopedia of Genes and Genomes; LGN, L451G452N453 amino acid sequence; MH, Mueller-Hinton; MIC, minimum inhibitory concentration; RMSD, root mean square deviation; RNAP, RNA polymerase; RRDRs, rifampicin resistance determining regions; SFA, saturated fatty acid; SNPs, single nucleotide polymorphisms; TCA, tricarboxylic acid; TF, trigger factor; UFA, unsaturated fatty acid.

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1 | Introduction

Molecular chaperones are vital components in prokaryotes and eukaryotes cells that assist in folding proteins into their native states. In *Escherichia coli*, trigger factor (TF), DnaK, and DnaJ (DnaK/J) perform molecular chaperones functions effectively. TF is the first chaperone encountered by nascent peptides from ribosomal exit, preventing them from aggregation and assisting them into correct conformation [1–4]. In addition to its cytoplasmic chaperone role, TF is responsible for sorting secretory proteins co-/posttranslationally [5–7]. Along with in-depth research, it is clearer that, TF as a secretory chaperone, links the cytoplasmic folding to secretion pathway chaperones by interacting with chaperone SecB and some translocases [8]. DnaK/J, together with GrpE, function as downstream of TF, assisting with protein folding, transporting, and disaggregation [9, 10]. The DnaK/J chaperone system acts like a “central hub” in *E. coli*, taking over the folding initiated by TF or transferring them to the GroEL/ES chaperone system [10]. Despite being at different positions within the *E. coli* chaperone system, TF and DnaK/J are crucial for maintaining proteostasis together with other chaperones and protease in *E. coli*. As TF and DnaK/J possess overlapping functions on sharing a substrate pool, their individual or combined absence impacts on cell viability differently. For instance, *E. coli* with either TF or DnaK/J deletions still survive at normal growth temperature of 37°C. However, with combined deletion of TF and DnaK/J, *E. coli* cannot survive at 37°C accompanying with large amount of protein aggregation in vivo [9, 11, 12].

E. coli RNA polymerase (RNAP) is a multisubunit enzyme that mediates gene expression and almost controls transcription of all the genes in vivo. It exists in two forms, a holoenzyme ($\alpha_2\beta'\omega$) that initiates transcription by recognizing the promoter, and a core enzyme ($\alpha^2\beta\beta'\omega$) that undertakes transcription elongation and termination after releasing σ subunit [13, 14]. By targeting and binding with the beta (β) subunit of RNAP, the antibiotics rifampicin inhibits transcription, prevents RNA and protein synthesis in bacteria, and aborts bacterial growth [15]. The β subunit of *E. coli* RNAP encoded by *rpoB* contains the rifampicin-binding domain, where residues 146, 511, 513, 529, 531, 533–534, 568, and 572 directly contact with rifampicin naphthyl moiety, simultaneously residues 143, 510–512, 514, 516, 525–526, 564, and 761 contact with rifampicin ansa moiety [16]. Apart from rifampicin-binding sites, the mutation in rifampicin-resistance-determining regions (RRDRs) in RpoB, including residues 507–533, residues 563–572, residue 146 and 687, were also closely related with presence of rifampicin-resistant strains [17, 18]. Therefore, the structural destabilization of RNAP from mutations in critical sites of RpoB may decrease the affinity of rifampicin for RNAP in prokaryotes, leading to rifampicin resistance.

In constructing *E. coli* without TF and DnaK/J, the typical *E. coli* mutant Z629 surviving at 30°C was generated, while one peculiar *E. coli* mutant Z625 surviving at 37°C was also generated. We compared fatty acid (FA) composition and susceptibility to rifampicin of *E. coli* mutants with wild-type *E. coli*. Then, we performed whole genomic sequencing and transcriptomic analysis. Our research aimed to investigate the difference in genetic background between *E. coli* mutants Z625

and wild-type *E. coli*, and to analyze the survival reason for the unique *E. coli* mutant Z625 with deletions of DnaK/J and TF at 37°C.

2 | Materials and Methods

2.1 | Bacterial Strains and Culture Conditions

The genes *tig*, *dnaK*, and *dnaJ* encoded chaperone TF, DnaK, and DnaJ in wild-type *E. coli* BW25113. The *E. coli* Z116 (Δ *tig::cat*, *Amp*^r) with deletion of TF, and *E. coli* Z625 and Z629 (Δ *tig::cat*, Δ *dnaK/dnaJ::kan*) with deletions of TF and DnaK/J were constructed as previously described using the phage lambda Red recombinase system [19, 20]. Briefly, the primer pair K-KJ1/K-KJ2 whose 5' region was homologous to the flanking regions of *dnaK/dnaJ*, was used to amplify a kanamycin resistance gene (*kan*) from the pKD4 plasmid. *E. coli* Z116, harboring the vector pKD46, was then transformed with the PCR products by electroporation to replace its *dnaK/dnaJ*. The *kan*⁺ and *amp*⁺ clones were screened on LB plates containing kanamycin and ampicillin, and characterized by PCR using the primer pair GKJ1/GKJ2, which targeted the upstream and downstream regions of the *dnaK/dnaJ* coding region, respectively. A 980-bp PCR product in 1% agarose gel was observed, indicating that *dnaK/dnaJ* was replaced by *kan*. The resultant strains were further confirmed by Western blot analysis using an anti-DnaK monoclonal antibody. No band was observed for in the *dnaK* clean deletion mutant. In the resultant strains with deletion of *tig* and *dnaK/J*, those growing at 30°C were named as Z629; conversely, the strains growing at 37°C were named as Z625. Both Z625 and Z629, containing the plasmid pKD46, were cultivated on LB plates containing kanamycin, chloramphenicol, and ampicillin at 37°C and 30°C. Next, the pKD46 plasmid was removed from Z625 and Z629 by culturing multiple generations of bacteria on LB plates containing kanamycin and chloramphenicol without ampicillin at 37°C and 30°C for 16 h. Then presence of pKD46 was confirmed by PCR using primers p46-1 and p46-2, which could amplify an 1150-bp region of pKD46; if pKD46 was absent, the 1150-bp fragment could not be amplified. The construction process of the mutants Z625 and Z629 was shown in Supporting Information S1: Figure S1. Besides genome sequencing, the deletions of *tig* and *dnaK/J* were also verified by PCR and western blot (Supporting Information S1: Figure S2). All the primers used are listed in Supporting Information S1: Table S2.

2.2 | Antimicrobial Susceptibility Tests

The minimum inhibitory concentration (MIC) of *E. coli* Z625, Z629, and wild-type BW25113 was determined using 25 mL of 90 mm Mueller–Hinton (MH) agar plates and commercially available *E*-test strips (Liofilchem SRL, Roseto degli Abruzzi, Italy) containing rifampicin at concentrations ranging from 0.016 to 256 μ g mL⁻¹. Overnight LB liquid containing chloramphenicol and kanamycin for Z625 and Z629, and LB liquid for BW25113, were used to suspend the bacteria. These suspensions were adjusted into a McFarland index of 0.5. A sterile swab was dipped into the bacteria suspensions and was

streaked over the entire surface of MH agar, following the instructions provided with the *E*-test strips. The MH plates, with an *E*-test strip attached, were then incubated at 30°C for Z629 and at 37°C for BW25113 and Z625. After 16 h, the MIC value was read at the point of complete inhibition. Each strain was tested in triplicate for accuracy.

2.3 | FA Methyl Ester (FAME) Analysis

FAs were extracted from BW25113, Z625, and Z629 using the ester-linked method. Initially, 40 mg of bacterial precipitate was mixed with 1 mL of a solution containing 3.75 M KOH in a methanol:distilled water reagent at a 1:1 ratio. The mixture was briefly vortexed and then heated in a boiling water bath for 30 min. After cooling, 2 mL of a 6.0 N hydrochloric acid:methyl alcohol solution at a 13:11 ratio was added to the samples, followed by further vortexing and heating for 10 min at 80°C. This step facilitated the release and methylation of ester-linked FAs from the microorganisms. Subsequently, FAMES were extracted into an organic phase consisting of hexane and hexane:methyl *tert*-butyl ether at a 1:1 ratio. The samples were then centrifuged at 480 g for 10 min to separate the aqueous and hexane phases, with the aqueous phase being discarded. The hexane layer was mixed with 3 mL of a 0.3 M KOH and tumbled for 5 min. Two-thirds of the resulting organic phase were pipetted into a gas chromatography system (Agilent 6890) to separate the individual FAMES. The temperature was programmed to increase from 170°C to 270°C at a rate of 5°C per min. Identification of FAs and determination of relative peak areas were performed using the MIDI Sherlock TSBA6 method and library. The experiments were repeated three times for consistency.

2.4 | Genomic Dna Preparation, DNA Library Construction, and Sequencing

Genomic DNA from *E. coli* Z625 and Z629 was extracted using the Qiagen DNeasy extraction kit, following the manufacturer's instructions. A concentration of $\geq 60 \text{ ng } \mu\text{L}^{-1}$ genomic DNA from each sample was used for subsequent sample preparations.

The bacterial genomic DNA was sequenced at BGI Genomics (Shenzhen, China) using two technologies: DNBSEQ sequencing and PacBio sequencing. For DNBSEQ sequencing, the genomic DNA was fragmented by sonication and then end-polished, A-tailed, and ligated with a full-length adapter, followed by PCR amplification. The resulting PCR products were purified using the AMPure XP-Medium system and then were assessed using an Agilent 2100 Bioanalyzer. The purified double-stranded PCR products were heat-denatured and circularized into the single-strand circle DNA (ssCir DNA) in the presence of a splint oligo sequence to form the final library. The sequence libraries were evaluated for size and concentration using Agilent 2100 Bioanalyzer, and then sequenced. For PacBio sequencing, the genomic DNA was subjected to g-TUBE treatment to produce appropriately sized fragments, then ends of which were repaired and ligated to the connector of the hairpin structure to form a dumbbell-shaped structure known

as 'SMRTbell'. The annealed SMRTbell template was combined with the polymerase to produce the SMRTbell library for the final sequencing.

2.5 | Data Filtering, Assembly of Bacterial Genome and Bioinformatics Analysis

To prepare for downstream analyses, raw sequencing reads were first processed to remove adapter reads, duplication reads and low-quality sequences, resulting in a set of high-quality, clean reads. The genomes of Z625 and Z629 were sequenced and assembled utilizing the long-read sequencing capabilities of the PacBio platform. The assembled genome of Z625 and Z629 were then aligned with the *E. coli* K-12. BW25113, which is referenced in the NCBI database under the RefSeq assembly accession number GCF_000750565.1. The comparison, facilitated by the NUCmer tool, revealed that a 9-bp nucleotide sequences insertion occurred within *rpoB* of Z625.

The I-Tasser server was used to obtain structural models of RpoB with insertion [21]. Pymol was used for structural visualization, by which protein models of RpoB with insertion was superimposed with wild-type RpoB in BW25113 (PDB:5UAC) to determine the structural changes at the insertion sites [22]. The structural similarity between two protein structures was assessed using root means square deviation (RMSD) calculated based on corresponding atoms of two protein chains [23].

2.6 | Microarray Operation and Data Analysis

The microarray experiments were carried out using the procedures and reagents as previously described [24]. Briefly, cDNA was synthesized from the total RNA isolated from fresh bacteria with an OD₆₀₀ of 0.6. The cDNA from BW25113 (at 30°C or 37°C), Z625 (at 30°C or 37°C) and Z629 (at 30°C) were labeled with Cy5- and Cy3-dCTP fluorescent dye, respectively. Four groups of chips were prepared: Z625 and Z629 cultured at 30°C (Comparison 1: Z625_Z629 at 30°C), BW25113 and Z629 cultured at 30°C (Comparison 2: BW25113_Z629 at 30°C), Z625 cultured at 30°C and Z625 at 37°C (Comparison 3: Z625 at 30°C_Z625 at 37°C), BW25113 and Z625 at 37°C (Comparison 4: BW25113_Z625 at 37°C). Hybridization and scanning followed a standard protocol, and each microarray experiments was performed three times. In the original microarray data, genes that displayed statistically significant differential expression between the two conditions were identified based on the following criteria: an average fold change of ≥ 2 for upregulation or ≤ 0.5 for downregulation, and *q*-values of quality control $\leq 1\%$. Differential expressed genes (DEGs) were grouped and annotated according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway database (www.genome.jp/kegg) and gene ontology (GO) database (geneontology.org).

2.7 | Data Availability

The transcription data were deposited in the NCBI Gene Expression Omnibus (GEO) database under platform identification number

GPL8850 and assigned the following accession number GSM1828833 for Comparison 1 (Z625_Z629 at 30°C), GSM1828837 for Comparison 2 (BW25113_Z629 at 30°C), GSM1828838 for Comparison 3 (Z625 at 30°C_Z629 at 37°C), and GSM1828835 for Comparison 4 (BW25113_Z625 at 37°C). The complete genomic sequencing data were deposited in the NCBI database under accession number PRJNA1017805.

3 | Results

3.1 | Z625 and Z629 Have Significant Difference in FA Composition

When exploring the effects of the triple deletion of the chaperones DnaK/J and TF on *E. coli* growth, we derived two mutants, Z625 and Z629. Both strains were created by replacing the *dnaK/J* and *tig* loci with kanamycin and chloramphenicol resistance genes, respectively, using the same deletion method. The accuracy of these genetic alterations was verified by PCR, and the absence of TF and DnaK/J expression was confirmed through Western blot analysis following our established methods [20]. Notably, Z625 was capable of surviving at 37°C, akin to the wild-type *E. coli* BW25113, while

Z629 was able to grow at temperature below 30°C. This resilience of Z625 is particularly remarkable as other *E. coli* strains lacking triple chaperones have not been reported to survive at 37°C in previous studies [11, 12, 25].

To determine if phenotypic difference exist in triple chaperones knockout strains with similar genetic background, we performed FAME analysis on Z625, Z629, and BW25113. The FA profiles of these three *E. coli* strains are displayed in Table 1. Relative to BW25113, the production of saturated FAs (SFAs) 17:0 cyclo in Z625 was significantly reduced, being 7.4 times lower, while the production of SFAs 16:0 was 1.3 times lower. Additionally, the production of SFAs 19:0 cyclo in Z625 was found to be negligible. In all three strains, several well-defined unsaturated FAs (UFAs) were detected only in trace amounts. Furthermore, the levels of mixed or identical UFAs 16:1 ω7c, 16:1 ω6c, 18:1 ω7c, and 18:1 ω6c, as shown in summed feature 2 and 3 in Table 1, were threefold higher in Z625 compared to both BW25113 and Z629. In summary, the FA profile of Z629 at 30°C closely resembled that of BW25113, but the profile of Z625 at 37°C was markedly different. Specifically, Z625 showed a significant increase in the production of UFAs and a decrease in SFAs.

TABLE 1 | Fatty acid composition of *Escherichia coli* BW25113, Z625 cultured at 37°C and Z629 cultured at 30°C.

	37°C BW25113	37°C Z625	30°C Z629
Saturated fatty acid			
12:0	3.69 ± 0.22	3.45 ± 0.53	3.83 ± 0.41
13:0	< 1	—	< 1
14:0	7.30 ± 0.64	6.45 ± 0.36	7.16 ± 0.81
15:0 iso	< 1	—	—
16:0	32.98 ± 1.1	24.74 ± 1.3	33.29 ± 0.9
17:0 cyclo	21.98 ± 0.72	2.97 ± 0.44	23.41 ± 1.21
17:0	< 1	—	< 1
18:0	< 1	< 1	< 1
19:0 iso	< 1	—	—
19:0 cyclo w8c	5.62 ± 0.22	< 1	5.17 ± 0.44
Unsaturated fatty acid			
16:1 ω9c	< 1	—	< 1
16:1 ω5c	< 1	< 1	< 1
18:1 ω5c	—	—	< 1
18:1 ω9c	—	—	< 1
18:1 ω7c 11-methyl	< 1	—	—
Summed features ^a			
Summed feature 1 ^b	7.54 ± 0.42	7.32 ± 0.59	7.65 ± 0.67
Summed feature 2 ^c	6.74 ± 0.37	18.75 ± 1.07	5.78 ± 0.57
Summed feature 3 ^d	11.83 ± 0.31	34.99 ± 1.93	11.39 ± 0.62

Note: Values given were percentages of the total fatty acids, which were means with standard error from three independent determinations. —, not detected; < 1, trace amount.

^aSummed features comprise a group of two or more fatty acids that could not be separated under the given chromatographic conditions using the MIDI system.

^b12:0 aldehyde, unknown 10.928, 16:1 iso I/14:0 3OH. The word “unknown” represent compounds which are not listed in the peak naming table of the MIDI system, and are therefore designated by their equivalent chain length.

^c16:1 ω7c and/or 16:1 ω6c.

^d18:1 ω7c and/or 18:1 ω6c.

3.2 | Whole-Genome Sequencing Identified a 9-Bases-Fragment Insertion Within *rpoB* of Z625

To investigate the genomic differences among Z625, Z629, and BW25113, we conducted third-generation PacBio sequencing in tandem with second-generation DNB sequencing. The dual sequencing strategy, known for its greater comprehensive and precise characterization of genomic structural variations such as copy-number variations, mobile element insertions, deletions, and translocations, was utilized for the whole-genome sequencing of Z625 grown at 37°C and Z629 grown at 30°C.

PacBio sequencing yielded a total of 485,032 subreads for Z625 and 536,918 subreads for Z629, averaging 917.61 base pairs in subread length. DNBSEQ sequencing generated 1302 megabases (Mb) of clean, paired-end 150-bp data for Z625, and 1303 Mb for Z629. Genome assembly details for Z625 and Z629 are available in Supporting Information S1: Table S1. Both assemblies showed identical counts of genes, coding sequences (CDS), RNA genes, and CRISPR arrays, indicating that gene content and numbers were not directly influencing the differential maximum growth temperatures of Z625 and Z629. Comparative genome analysis between Z625 and Z629 revealed a 9-base pair insertion duplication in Z625, specifically “CTCGGCAAC.” This insertion led to an additional amino acid sequence L451G452N453 (LGN) in the *RpoB* protein of Z625. In contrast, the *RpoB* sequence in Z629 was consistent with that of BW25113 (Supporting Information S1: Figure S3). Additionally, comparisons of the whole genomes of both Z625 and Z629 to BW25113 confirmed that the *tig* gene was replaced by chloramphenicol resistance gene (*chl*) and the *dnaK* and *dnaJ* genes were replaced by a kanamycin resistance gene (*kana*). The whole-genome sequencing thus verified the successful knockout of triple chaperone genes in Z625 and Z629. However, it is important to note that the 9-base pair insertion is not the only change, as additional variations in the genomes of both Z625 and Z629 were observed comparing to BW25113. Three shared single nucleotide polymorphisms (SNPs) were identified in Z625 and Z629, “T-C” at position 1092433, “C-A” at position 1092574, and “C-T” at position 1092614, all within an intergenic spacer between *ycdU* and *ghrA*. The “wild type” sequence (“T” at position 1092433, “C” at 1092574 and “C” at 1092614, according to *E. coli* K-12 substr. BW25113 [GenBank: CP009273.1; NCBI Reference Sequence: NZ_CP009273.1]) could match those from many other *E. coli* strains, such as *E. coli* strain Z142-1 (GenBank: CP159700.1), *E. coli* strain Z217-7 (GenBank: CP159655.1), *E. coli* strain Ec0900 (GenBank: CP158187.1), *E. coli* strain KFu015 (GenBank: CP158187.1). However, “mutant” sequence (“C” at 1092433, “A” at 1092574 and “T” at 1092614, as seen in Z625 or Z629) also match *E. coli* strain EC-YC1908-399 (GenBank: CP084534.1) with 100% identities, and match many other *E. coli* strains with only one base mismatch. Therefore, these SNPs, or their combinations, are widely present in the numerous different *E. coli* strains. However, as these SNPs are situated in intergenic regions and do not encode any proteins or functional RNAs, they are unlikely to impact the biological functions of the strains. Through genome resequencing, we observed that genes knockout perfectly matched our expectations without any

off-target occurrences. This verifies the exceptionally high reliability of the λ -Red recombineering system [19, 20].

3.3 | Z625 and Z629 Exhibited Identical Susceptibility to Rifampicin

Considering that *RpoB* is the target protein with which rifampicin interacts, we assessed the influence of the LGN insertion in *RpoB* on *E. coli*'s susceptibility to rifampicin by using the *E*-test method to determine the MIC. As depicted in Figure 1, all three strains (BW25113, Z625, Z629) showed an identical MIC value of 4 $\mu\text{g/mL}$ for rifampicin. According to the susceptibility interpretive criteria provided in the handbook, an MIC value of $\leq 4 \mu\text{g/mL}$ for *Enterobacteriaceae* suggests susceptibility to rifampicin. Therefore, both mutant strains Z625 and Z629, as well as reference strain BW25113, were deemed susceptible to rifampicin and were not considered resistant.

Since LGN insertion did not induce resistance of Z625 to rifampicin, it was speculated that location of LGN in space were not closely related to rifampicin binding sites in *RpoB*. Next, the protein sequence of *RpoB* with the LGN insertion from Z625 was modeled into the crystal structures of the *E. coli* *RpoB* using the Iterative Threading Assembly Refinement (I-TASSER) online server, aiming to simulate the structure of the β subunit of RNAP with the LGN insertion and to detect specific location if the LGN insertion in space. After achieving the resulting PDB file of β subunit with LGN insertion, the predicted structure of this modified β subunit was then compared to that of the wild-type *E. coli* *RpoB* (Supporting Information S1: Figure S4). A relatively low RMSD value of 0.054, calculated with PyMOL software, indicated a minimal structural divergence between the *RpoB* with the insertion and the wild-type *RpoB*. In the modeled structure, the LGN insertion area did not overlap with any of the rifampicin binding sites or the resistance-determining regions. Instead, it extended away from the core region of the *RpoB*, that interacts with rifampicin (Supporting Information S1: Figure S4). Given that rifampicin inhibits bacterial transcription by binding to *RpoB*, changes within the *rpoB* gene could potentially affect the interaction [15]. The structural modeling suggested that the LGN insertion did not induce significant structural changes to *RpoB*, in the meantime, did not disrupt rifampicin binding sites in Z625. This conclusion supported by *E*-test results showing that, Z625, like Z629 and BW25113, remained susceptible to rifampicin. However, the whole RNA polymerase carrying the *RpoB* with LGN insertion, probably have a potential effect on transcriptional

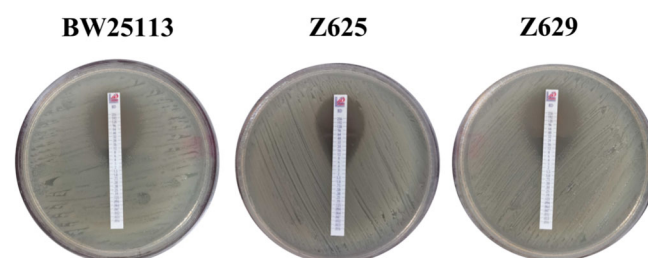


FIGURE 1 | Rifampicin susceptibility of BW25113, Z625, and Z629 determined by *E*-test.

expression of genes in Z625, which would be investigated by transcriptomic analysis.

3.4 | Insertion in *rpoB*, Deletion of Chaperone and Temperature Induced Different Gene Transcription

Compared to Z629, the only genomic difference observed in Z625 is a 9-base pair insertion in the *rpoB*. Although this insertion does not disturb RpoB of Z625 to bind with rifampicin, it was speculated that RpoB of Z625 may still impact on transcriptional function of the intact RNAP, resulting in transcriptional changes of multiple genes and subsequently affecting multiple metabolic pathways in Z625. Therefore, we conducted four sets of transcriptomic comparisons involving BW25113, Z625, and Z629 at their respective cultivation temperatures to investigate the transcriptomic changes associated with viability of Z625 at 37°C (Figure 2A). Table S3 containing a list of DEGs from the four comparisons, including their gene names, symbols, fold changes, and functional descriptions. From the 4 sets of comparisons, it is evident that three factors, molecular chaperone deletion, cultivation temperature, and the insertion in *rpoB*, impact the expression profile of *E. coli*. To further distinguish which genes were affected by the aforementioned factors in *E. coli*, we categorized the affected genes into four classes (Figure 2B).

A Venn diagram revealed overall status of co-expressed gene in four comparisons, indicating that a substantial overlap with most significant number of DEGs (153) were shared between Comparison 4 and Comparison 3, and Comparison 1 were shared a lesser degree of commonality with the DEGs from other comparisons (Figure 3A). GO enrichment revealed distinct enrichment patterns of DEGs across Class 1–4, which were affected by individual factor or two combined factors. (Figure 3B–F). Class 1 genes showed significant enrichment in “protein folding,” “response to temperature stimulus,” and “response to heat.” This class included chaperone genes such as *groEL/ES*, *secB*, *ibpA/B*, protease genes like *clpB*, *hslO/R/V/U*, and sigma factor *rpoD* (Figure 3C). The transcriptional alterations in these genes were predominantly induced by the deletion of DnaK/J and TF, manifesting as upregulation in Z625 or Z629 relative to BW25113. Additionally, genes associated with “maltose transport” and “carbohydrate transport” were enriched in Class 1, indicating that their transcriptional change was also induced by the deletion of DnaK/J and TF.

Class 2 genes were enriched in biological processes related to respiration, including cellular and anaerobic respiration (Figure 3D), suggesting a response to higher cultivation temperature or insertion in *rpoB* at 37°C. A distinct enrichment in genes associated with aerobic respiration was observed in Class 3, implying an impact from the insertion within *rpoB* (Figure 3E). Class 4 not only shared overlaps with Class 3 in the “oxidation reduction process,” but also exhibited significant enrichment in “fatty acid metabolic processes,” and “ribosome assembly.” This class also additionally highlights the respiration process, encompassing genes involved in both aerobic and anaerobic respiration (Figure 3F), suggesting a response to the insertion in *rpoB* at 37°C.

In summary, the enriched biological processes identified in Class 1 DEGs are distinctly different from those in Class 2, 3, and 4, thereby indicating that chaperone deletion and *rpoB* insertion exert divergent effects on the transcriptional regulation of different sets of genes.

3.5 | Transcriptional Analysis of Genes Related With Viability of Z625 at 37°C

To elucidate the mechanisms behind the survival of Z625 and the lethality of Z629 at 37°C in the absence of triple chaperones, we conducted a KEGG enrichment analysis comparing the expression levels of genes involved in protein, glucose, and lipid metabolism.

In the context of protein folding and degradation, cytoplasmic chaperone genes (*ibpA/B* and *groEL/S*) and protease genes (*clpB*, *hslU/V*) were upregulated in both Z629 at 30°C and Z625 at 37°C compared with BW25113. In contrast, periplasmic acid-resistance chaperone genes (*hdeA/B*) were only downregulated in Z625 compared with BW25113 at 37°C, as illustrated in Figure 4. These findings suggest that most of cytoplasmic folding modulators' genes are compensatory upregulation in response to the absence of major chaperones TF and DnaK/J in both Z629 and Z625, a pattern which closely resembles the early discovery of upregulation of cytoplasmic chaperone genes in absence of single chaperone TF or DnaK/J [24, 26]. Moreover, genes encoding RNAP coenzymes (*rpoB/C*) and one sigma factor gene (*rpoD*) were upregulated in Z625 compared with BW25113 at 37°C. Conversely, in Z629 compared to BW25113 at 30°C, only sigma factor genes (*rpoD/E/H* and *fecI*), but not any gene encoding core components of RNAP, were induced. Additionally, nearly all ribosomal protein genes and transcription elongation factors genes were consistently upregulated in Z625 compared to BW25113 at 37°C, while their expression remained stable in Z629 relative to BW25113 at 30°C (Figure 4). Furthermore, more DEGs were shown in Z625 than in Z629 (Figure 2A), indicating that insertion in *rpoB* of Z625 induced significant alterations in the transcriptional machinery of Z625.

In glycolysis, genes encoding enzymes essential for catalytic steps leading up to pyruvate production were downregulated in Z625 compared to BW25113 at 37°C (Figure 4). During the TCA cycle, where pyruvate is metabolized, nearly all related genes were upregulated in Z625 compared to BW25113 at 37°C. Moreover, an array of genes involved in aerobic respiration were upregulated in Z625 compared to BW25113 at 37°C, while genes associated with anaerobic respiration were downregulated (Figure 4), indicating a shift towards for aerobic respiration in Z625 at 37°C. In contrast, fewer genes in the TCA cycle and those associated with anaerobic and aerobic respiration were downregulated in Z629 compared to BW25113 at 30°C.

Analysis of FA composition revealed an increase in UFAs and a decrease in SFAs in Z625, prompting further investigation into genes related to FA biosynthesis and degradation. Although the genes *fabB* and *fabA*, which are pivotal in determining UFA content [27], and the regulator gene *fadR* (FA degradation) and

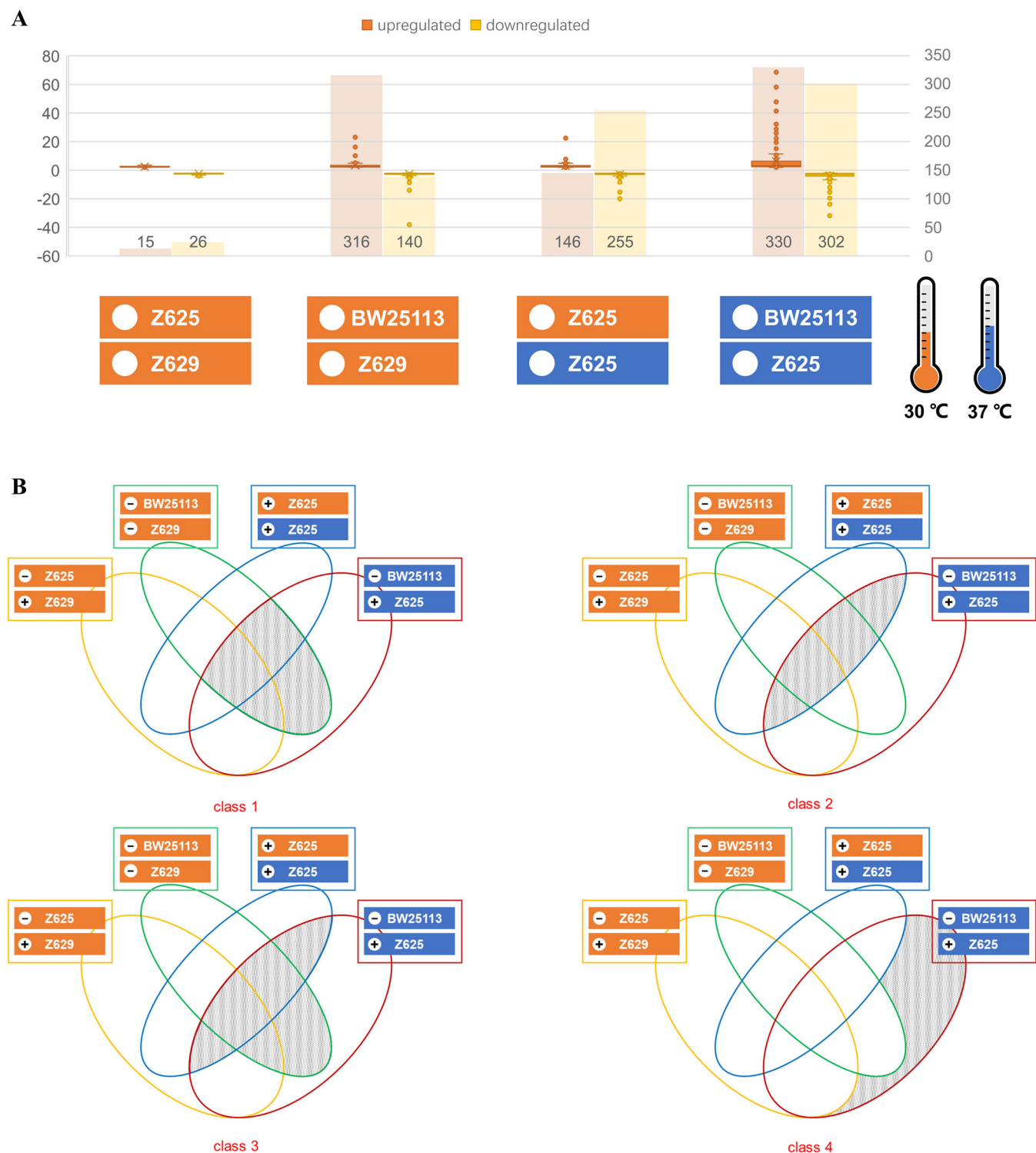


FIGURE 2 | Legend on next page.

fabR (FA biosynthesis) exhibited stable expression in Z625 compared to BW25113 at 37°C, genes involved in FA biosynthesis (*accA*) and degradation (*fadA/B/E/J*) were upregulated in Z625 (Figure 4). Upregulation of *accA* could potentially increase the supply of malonyl-CoA, a precursor for FA biosynthesis, while upregulation of *fadA/B/E/J* could contribute to more intensive FA degradation. These transcriptional changes likely contributed to the significant increase in UFAs (summed

features 2 and 3) and a decrease in SFAs (17:0 cyclo and 16:0), as detailed in Table 1.

In the transcriptomic data for Z629 compared to BW25113 at 30°C, only two FA degradation genes (*atoB/fadE*), along with one UFA synthesis gene (*fabA*), were upregulated (Figure 4). Relative to BW25113, traces of SFA (19:0 iso) and UFA (18:1 ω7c 11-methyl) completely disappeared in Z629, while new

traces of UFAs (18:1 ω 5c, 18:1 ω 9c) were produced (Table 1). Therefore, it was postulated that the upregulation of *fabA* was more conducive to the production of a greater variety of UFAs rather than an increase in their quantity. In comparison to Z625, fewer genes related to FA degradation were upregulated in Z629. Therefore, the degradation of SFAs and UFAs in Z629 was not as pronounced as in Z625.

4 | Discussion

Generally, viability of *E. coli* depletion of DnaK/J and TF is compromised under the regular cultivation temperature of 37°C. However, an unintended mutant *E. coli* Z625, with triple chaperones deleted, managed to survive at 37°C. By whole genome sequencing, a 9-bp repetitive sequence insertion was identified within the *rpoB* of Z625. Remarkably, this insertion, which conferred the ability to survive at 37°C, has been retained as a beneficial mutation in the Z625 genome.

Based on the simulated RpoB structure with the insertion and the observation that Z625 remains sensitive to rifampicin in E-test experiments, the insertion in Z625 does not disrupt RpoB to bind with rifampicin effectively. Nevertheless, the substantial presence of UFAs in Z625, as shown in FAME, as well as transcriptomic differences between Z625 and Z629, indicates that metabolic adjustments and alterations in membrane structural compositions resulting from the insertion in *rpoB* could explained the improved viability of Z625. We categorized DEGs into 4 classes induced by chaperone deletion, insertion in *rpoB* and/or higher cultivation temperature. The involved genes associating with viability of Z625 are depicted in Figure 5 to illustrate their participation in relevant biological pathways.

First, it was obviously that chaperones deletion induced compensatory upregulation of certain chaperone and protease genes (Figure 5), which could eliminate aggregated proteins resulting from the absence of TF and DnaK/J-assisted folding, thereby sustaining the viability of Z625 and Z629. In the case of Z625, which lacked the protection of triple molecular chaperones and thrived at higher culture temperatures 37°C, protein

aggregation within its cells may be even more severe compared to Z629 cultured at 30°C. The higher upregulation fold of chaperone *groEL/ES* and protease *hslUV* in Z625 compared to Z629 facilitates Z625 in attenuating protein aggregation at 37°C, thereby effectively maintaining proteostasis.

The number of downregulated genes (302) in Z625 compared to BW25113 at 37°C is higher than the downregulated genes (140) in Z629 compared to BW25113 at 30°C. Among the downregulated genes in Z625, the majority are involved in processes related to respiration, transport, and redox stress (Table S3). Most proteins encoded by these genes are membrane proteins, which are particularly prone to aggregation at higher temperatures. Therefore, we hypothesized that the insertion in *rpoB* induced downregulation of genes whose overexpression is toxic to *E. coli* in the absence of triple chaperones. The transcriptional repression resulting from insertion in *rpoB* partially alleviated the protein aggregation damage caused by protein overexpression at elevated cultivation temperatures, potentially explaining why Z625 with triple chaperones deletion can survive at 37°C.

In addition to the transcriptional suppression mentioned above, the survival of Z625 at 37°C is achieved from metabolic adjustments that lead to corresponding changes in membrane composition (Figure 5). FAME profiles revealed that Z625 has a higher concentration of UFAs, indicating that the cell membrane composition of Z625 is characterized by a high presence of UFAs. The upregulated *accA* can accelerate the production of precursor malonyl-CoA for FA, favoring the whole FA synthesis in Z625. The upregulated genes in aerobic respiration suggested an increased availability of acetyl-CoA and ATP, providing crucial precursors and meeting the additional energy demands for FA synthesis in Z625. The upregulated *icd* in Z625 could increase the supply of NADPH, that are essential reducing power for producing FAs. However, alongside various transcriptional changes in genes related to FA synthesis, a significant portion of genes related to FA degradation has also been upregulated in Z625 (Figure 5), suggesting that FA degradation was also active in Z625. Given that the substantial synthesis of FAs occurred in Z625 (Table 1), these changes are likely a response to maintain lipid balance. The specific reasons for the increased yield

FIGURE 2 | (A) The schematic diagram illustrated transcriptomic comparisons for BW25113, Z629, and Z625 cultured at 30°C or 37°C. Comparison 1 (Z625_Z629 at 30°C) demonstrated that the impact of the insertion in *rpoB* on the differentially expressed genes (DEGs) of *Escherichia coli* with deletion of DnaK/J and TF at 30°C. Comparison 2 (BW25113_Z629 at 30°C) demonstrated that the impact of deletion of DnaK/J and TF on DEGs of *E. coli* at 30°C. Comparison 3 (Z625 at 30°C_Z625 at 37°C) demonstrated that the impact of temperature (30°C and 37°C) on DEGs of *E. coli* with deletion of DnaK/J and TF and insertion in *rpoB*. Comparison 4 (BW25113_Z625 at 37°C) showed the combined impact of deletion of DnaK/J and TF and insertion in *rpoB* on DEGs of *E. coli* at 37°C. The right y-axis represented numbers of DEGs, while the left y-axis represented the expression fold of DEGs. Orange backgrounds in boxes indicated bacteria cultivated at 30°C, and blue backgrounds indicated bacteria cultivated at 37°C. The strain names were marked within each box. Symbol “+” denoted presence of a 9-bp insertion in *rpoB*, while “-” denoted a normal *rpoB*. The numbers represented upregulated or downregulated genes. (B) The schematic diagram represented four classes of genes induced by temperature, insertion in *rpoB* or chaperone deletion. Class 1 genes demonstrated that genes exhibit the identical transcriptional change, that is, upregulation or downregulation in both Comparison 2 and Comparison 4, indicating that their transcriptional changes are induced by the deletion of DnaK/J and TF, not induced by the insertion in *rpoB*. Class 2 genes demonstrated that genes exhibited identical transcriptional change, that is, upregulation or downregulation in both Comparison 3 and Comparison 4, indicating that their transcriptional changes are induced by higher cultivation temperature (37°C), or by insertion in *rpoB* and cultivation factor 37°C. Class 3 genes demonstrated that genes exhibited different transcriptional between Comparison 2 and Comparison 4, or between Comparison 3 and Comparison 4, that is, upregulation in Comparison 2 and downregulation in Comparison 4 and vice versa, or upregulation in Comparison 3 and downregulation in Comparison 4 and vice versa, indicating that their transcriptional changes are induced by insertion in *rpoB*. Class 4 genes demonstrated that genes exhibited transcriptional changes solely in Comparison 4, indicating that their transcriptional changes are induced by insertion in *rpoB* and higher cultivation temperature (37°C).

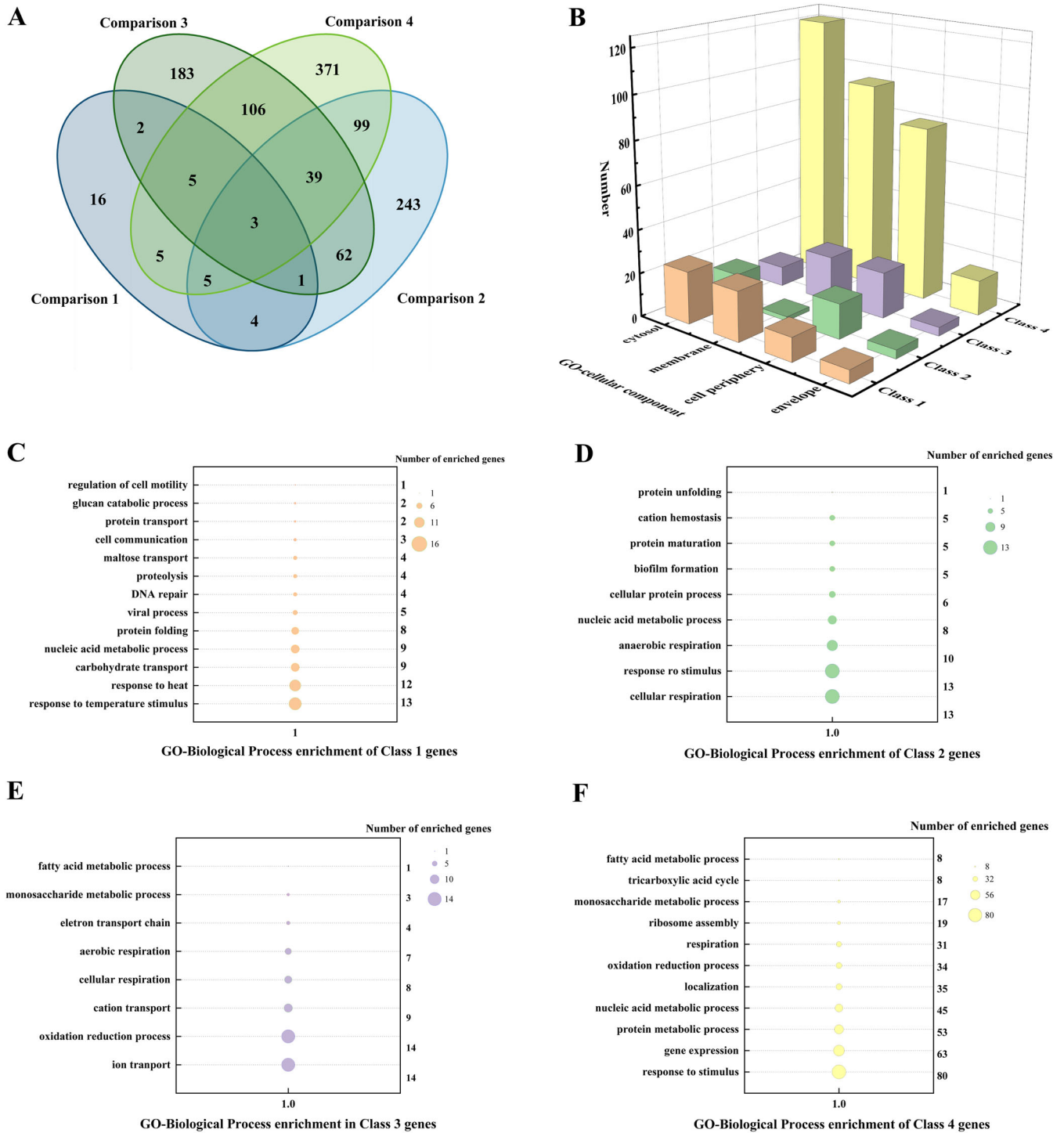


FIGURE 3 | The transcriptomic analysis elucidated the differential expression genes (DEGs) associated with the distinct genetic modifications across the four comparisons (A) Venn diagram of DEGs demonstrated the intersection of DEGs across Comparison 1, Comparison 2, Comparison 3, and Comparison 4. (B) GO-cellular components analysis for DEGs from class 1, class 2, class 3, and class 4. (C–F) GO-biological process analysis for DEGs from class 1, class 2, class 3, and class 4.

of UFAs in Z625 remains unclear. Nevertheless, the increased presence of UFAs in the membrane of Z625 can alter cell membrane fluidity and reduce the detrimental effects of high temperature on cell membrane function, enhancing Z625's adaptation to higher cultivation temperatures.

Almost every gene encoding key enzymes in glycolysis exhibited downregulation, suggesting that glycolysis in Z625 was

reduced. Furthermore, the upregulated genes in aerobic and TCA cycles, along with downregulated genes in anaerobic respiration (Figure 5), suggests that aerobic respiration remained the preferred metabolic pathway in Z625. This enhanced respiration efficiency facilitated the effective degradation of excess FAs and amino acids generated from the breakdown of aggregated proteins, particularly under conditions of active FA synthesis and degradation of misfolded proteins in Z625. Thus, the

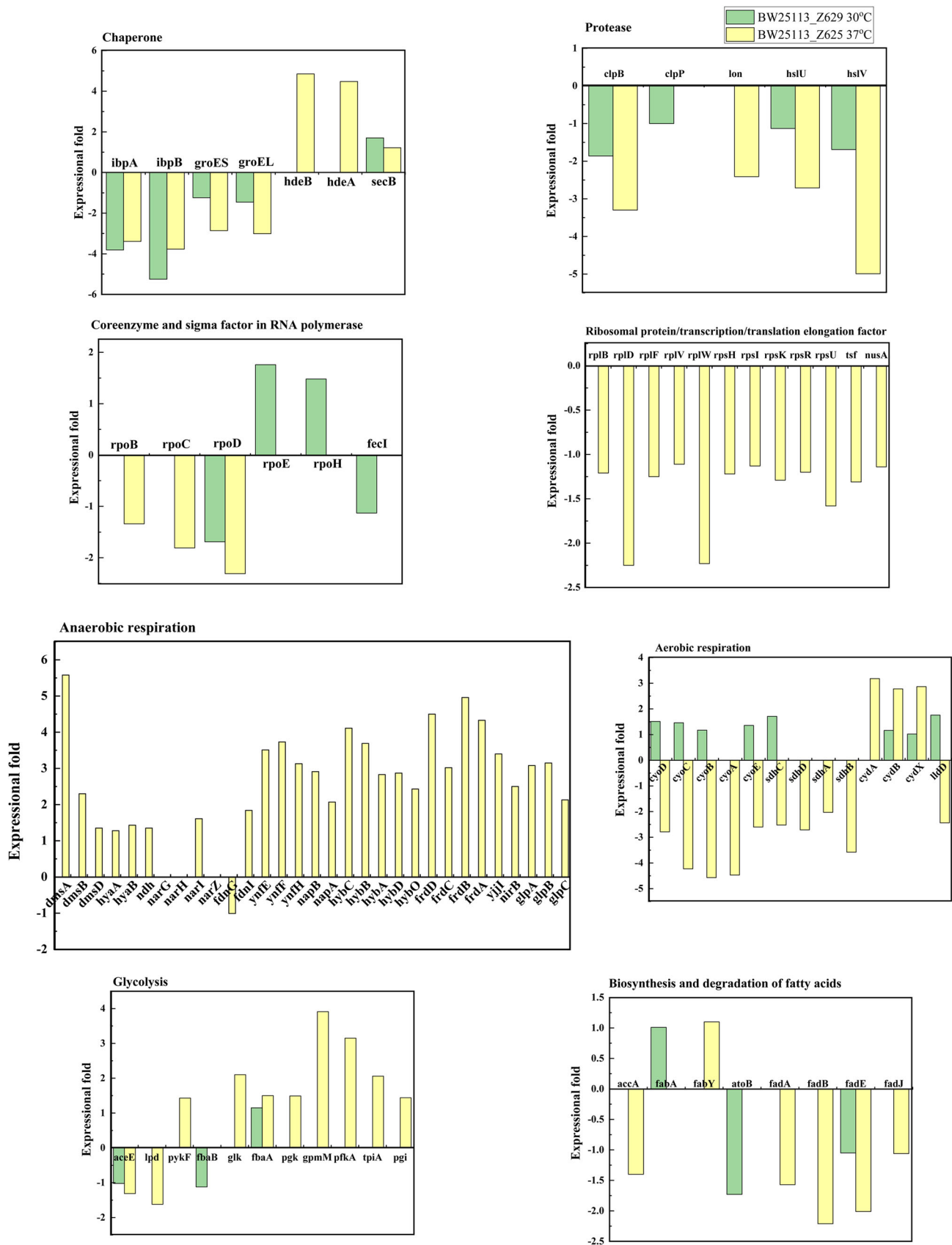


FIGURE 4 | Expression changes of representative DEGs in the KEGG pathway from Comparison 2 (BW25113_Z629 at 30°C) and Comparison 4 (BW25113_Z625 at 37°C). Negative expression fold indicated gene upregulation in mutant, while positive expression fold indicated gene downregulation in mutant compared to BW25113.

upregulated aerobic respiration pathway and TCA cycle could still receive pyruvate inflows from catabolism of amino acids derived from degraded misfolded proteins. The fermentation process was attenuated by upregulated aerobic respiration and TCA cycle, resulting in a reduction of lactic acid production, which failed to accelerate the intracellular pH. The down-regulation of acid stress chaperone gene *hdeA/B* also indicated that there was no requirement to increase intracellular pH to activate *hdeA/B* to assist protein folding in Z625 (Figure 5).

Prior investigations have demonstrated that *rpoB* mutations can compensate for deficiencies resulting from mutations in specific genes of *E. coli* [31]. Additionally, owing to the frequent occurrence of epistatic interactions between *rpoB* and the rest of the genome, mutations like *rpoB* have substantial impacts on global transcriptional responses and play a crucial role in the evolution of host organisms [32]. Notably, when *E. coli* was subjected to thermal stress and evolved at 42°C across 114 separate populations, all

In summary, the unexpected LGN insertion in *rpoB* allowed Z625 to develop its own unique metabolic pattern, enabling survival at 37°C. For instance, upregulation of chaperones and proteases helped attenuate the aggregated proteins in the absence of DnaK/J and TF. The unexpected insertion into *rpoB* repressed the transcription of certain genes encoding proteins whose overexpression would be toxic to *E. coli*. Overall transcriptomic adjustments in FA biosynthesis pathway led to enhancements in UFAs, which improved membrane fluidity and contributed to survival at 37°C. The upregulated aerobic respiration and TCA cycle are advantageous for increasing catabolism of overproduced FAs and amino acids from degraded proteins, leading to lactate attenuation in Z625 and better adaption to elevated temperatures. The simple mutations in critical genes like *rpoB* might play an important role in driving adaptive evolution. With the continual advancements in gene sequencing technologies, it is ever more important to understand the patterns and mechanisms of hotspot mutations in critical genes like *rpoB*, which holds the potential to provide valuable insights into the molecular mechanisms that drive evolution.

Dongjie Fan: formal analysis, investigation, methodology, writing—original draft, writing—review and editing. **Lushan Liu:** formal analysis,

investigation. **Bella Yuen:** investigation, writing–review and editing. **Lu Sun:** formal analysis. **Rui Liao:** formal analysis. **Yanli Qu:** formal analysis. **Chuanpeng Liu:** writing–review and editing, funding acquisition. **Qiming Zhou:** writing–review and editing, project administration, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in NCBI BioProject Submissions at <https://submit.ncbi.nlm.nih.gov/subs/bioproject/>, reference number PRJNA1017805. All data generated or analyzed during this study are included in this article and its supplementary material files. Further enquiries can be directed to the corresponding author.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.